REVISION COM110

1. Given that $\int_1^{10} f(x)\,dx = 16$, $\int_{10}^{20} f(x)\,dx = 5$ and $\int_{-5}^{20} f(x)\,dx = -10$, determine the value of the following:

   (a) $\int_1^{20} f(x)\,dx$

   (b) $\int_{10}^{-5} f(x)\,dx$

   (c) $\int_{-5}^{1} f(x)\,dx$

2. Show that $y = 6 - \frac{A}{x^2}$ is a solution of the differential equation $2x\frac{dy}{dx} + 4y = 24$, where A is a constant.

3. The world population at the beginning of 1980 was 4.5 billion. Assuming that the population continued to grow at the rate of approximately 2% per year, the population growth can be modelled by the differential equation $\frac{dQ}{dt} = 0.02Q$, where $Q(t)$ represents the population (in billions) at time (in years, with $t = 0$ corresponding to 1980).

   (a) Prove that the general solution to this differential equation is $Q(t) = 4.5e^{0.02t}$.

   (b) Using the result from (a), what will be the world population at the beginning of 2019. Round your final answer to four decimal places (billions).

4. One model for the spread of a rumor is that the rate of spread is proportional to the product of the fraction P of the population who have heard the rumor and the fraction who have not heard the rumor. Hence, the model can be expressed by the differential equation $\frac{dP}{dt} = kP(1-P)$, where k is the rate of spread of the rumor.

   (a) Prove that $P(t) = \frac{1}{1 + Ae^{-kt}}$.

   (b) A small town has 1000 inhabitants. At 8 am, 80 have heard a rumor. Given $k = 0.6106$, at what time will 90% of the population have heard the rumor.

1. Given that $\int_1^{10} f(x)\,dx = 16$, $\int_{10}^{20} f(x)\,dx = 5$ and $\int_{-5}^{20} f(x)\,dx = -10$, determine the value of the following:

(a) $\int_1^{20} f(x)\,dx$

(b) $\int_{10}^{-5} f(x)\,dx$

(c) $\int_{-5}^{1} f(x)\,dx$

- $\int_a^a f(x)dx = 0.$
- $\int_a^c f(x)dx + \int_c^b f(x)dx = \int_a^b f(x)dx.$ 当 c 的位置无关 (as long as integrable)
  - $a<c<b$ ①
  - $a<b<c$ ②
- $\int_a^b f(x)dx = -\int_b^a f(x)dx.$

a) $\int_1^{20} f(x)\,dx = \int_1^{10} f(x)\,dx + \int_{10}^{20} f(x)\,dx = 16 + 5 = 21$

b) $\int_{10}^{-5} f(x)\,dx = \int_{10}^{20} f(x)\,dx + \int_{20}^{-5} f(x)\,dx$

$= 5 - \int_{-5}^{20} f(x)\,dx = 5 - (-10) = 15$

c) $\int_{-5}^{1} f(x)\,dx = \int_{-5}^{10} f(x)\,dx + \int_{10}^{1} f(x)\,dx$

$= 15 + 16 = -31$

2. Show that $y = 6 - \frac{A}{x^2}$ is a solution of the differential equation $2x\frac{dy}{dx} + 4y = 24$, where A is a constant.

$y' = +2Ax^{-3}$

$2x \cdot 2A\frac{1}{x^3} + 4\left(6 - \frac{A}{x^2}\right)$

$= \frac{4A}{x^2} + 24 - \frac{4A}{x^2} = 24$

LHS of DE = RHS of DE

$\therefore$ $y = 6 - \frac{A}{x^2}$ is a solution of the DE

let time t

3. The world population at the beginning of 1980 was 4.5 billion. Assuming that the population continued to grow at the rate of approximately 2% per year, the population growth can be modelled by the differential equation $\frac{dQ}{dt} = 0.02Q$, where $Q(t)$ represents the population (in billions) at time (in years, with $t = 0$ corresponding to 1980).

separable $\frac{1}{Q}dQ = 0.02dt$

(a) Prove that the general solution to this differential equation is $Q(t) = 4.5e^{0.02t}$.

(b) Using the result from (a), what will be the world population at the beginning of 2019. Round your final answer to four decimal places (billions).

a) $\frac{dQ}{dt} = 0.02Q$

$\frac{1}{Q}dQ = 0.02\,dt$

$\ln|Q| = 0.02t + C$

$|Q| = e^{0.02t+C}$

$Q = Ae^{0.02t}$, where $A = \pm e^{C}$

$Q(0) = A = 4.5$

$\therefore Q = 4.5e^{0.02t}$ ✓

b) $t = 2019 - 1980 = 39$

$Q(39) = 4.5e^{0.02\times 39} = 9.8166$ (billions) ✓

4. One model for the spread of a rumor is that the rate of spread is proportional to the product of the fraction P of the population who have heard the rumor and the fraction who have not heard the rumor. Hence, the model can be expressed by the differential equation $\frac{dP}{dt} = kP(1-P)$, where k is the rate of spread of the rumor.

(a) Prove that $P(t) = \frac{1}{1+Ae^{-kt}}$.

(b) A small town has 1000 inhabitants. At 8 am, 80 have heard a rumor. Given $k = 0.6106$, at what time will 90% of the population have heard the rumor.

(a) $\frac{dp}{dt} = kp(1-p)$

$\int \frac{1}{p(1-p)}\,dp = k\cdot\int dt$

$\int \frac{A}{p} - \frac{B}{1-p}\,dp = kt + C$

$A(1-p) - Bp = 1$

① when $p = 0$

$A = 1$

② when $p = 1$

$-B = 1 \Rightarrow B = -1$

$\int \frac{1}{p} + \frac{1}{1-p}\,dp = kt + C$

$u = 1-p;\ dp = -du$

$\ln|p| - \ln|1-p| = kt + C$

$\ln\left|\frac{p}{1-p}\right| = kt + C$

$\frac{1-p}{p} \Rightarrow \frac{1}{p} - 1$

$\left|\frac{p}{1-p}\right| = e^{kt+C}$

$\frac{p}{1-p} = Ae^{kt}$, where $A = \pm e^{C}$

$\frac{1-p}{p} = \frac{1}{Ae^{kt}}$

$\frac{1}{p} = \frac{1}{Ae^{kt}} + 1$

$\frac{1}{p} = \frac{1+Ae^{kt}}{Ae^{kt}}$

$p = \frac{Ae^{kt}}{1+Ae^{kt}}$

$p = \frac{1}{\frac{1}{Ae^{kt}} + 1}$

$p = \frac{1}{Be^{-kt}+1}$, when $B = \frac{1}{A}$

$\ln\left|\frac{1-p}{p}\right| = -kt - C$

or $\ln\left|\frac{P}{1-P}\right| = kt + C$

$\ln\left|\frac{1-P}{P}\right| = -kt - C$

$\left|\frac{1-P}{P}\right| = e^{-kt-C}$

$\frac{1-P}{P} = Ae^{-kt}$, where $A = \pm e^{-C}$

$\frac{1}{P} = Ae^{-kt} + 1$

$P = \frac{1}{Ae^{-kt} + 1}$

4. One model for the spread of a rumor is that the rate of spread is proportional to the product of the fraction P of the population who have heard the rumor and the fraction who have not heard the rumor. Hence, the model can be expressed by the differential equation $\frac{dP}{dt} = kP(1-P)$, where k is the rate of spread of the rumor.

(a) Prove that $P(t) = \frac{1}{1+Ae^{-kt}}$.

(b) A small town has 1000 inhabitants. At 8 am, 80 have heard a rumor. Given $k = 0.6106$, at what time will 90% of the population have heard the rumor.

5. Solve the ordinary differential equation $y''+3y'+2y=x^2+5$ by the method of undetermined coefficients.

6. Solve the initial value problem $y''-4y'-12y=3e^{5t}, y(0)=\frac{18}{7}, y'(0)=\frac{-1}{7}$.

7. Evaluate the following indefinite integral:

   (a) $\int x^5+2x\,dx$

   (b) $\int \frac{3}{2x}\,dx$

   (c) $\int \frac{x^5+x+5}{x^2}\,dx$

   (d) $\int \sin^2(x)\cos^5(x)\,dx$

8. Evaluate the definite integral $\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{\cos x}{\sin^2 x}\,dx$. Round your final answer to four decimal places.

9. Use an appropriate substitution to solve the indefinite integral $\int 2x^2 e^{x^3}\,dx$.

10. Use an integration by parts to solve the indefinite integral $\int 2x\cos(5x)\,dx$.

11. Consider a rational function, $f(x)=\frac{x+11}{x^2+x-6}$.

    (a) Perform partial fraction decomposition on the given rational function.

    (b) Using the partial fraction decomposition obtained in (a), evaluate the definite integral $\int_5^6 f(x)\,dx$. Round your final answer to four decimal places.

12. Use a trigonometric substitution to solve the indefinite integral $\int \frac{\sqrt{16-x^2}}{x^2}\,dx$.

5. Solve the ordinary differential equation $y''+3y'+2y=x^2+5$ by the method of undetermined coefficients.

① y_c

let $y_c = e^{rx}$

$y' = re^{rx}$

$y'' = r^2e^{rx}$

$y'' + 3y' + 2y_c = 0$

$r^2e^{rx} + 3re^{rx} + 2e^{rx} = 0$

$(r^2 + 3r + 2)e^{rx} = 0$

$e^{rx} = 0$ or $r^2+3r+2 = 0$

(impossible) $(r+2)(r+1) = 0$

$r_1 = -2$ or $r_1 = -1$

$y_c = c_1e^{-2x} + c_2e^{-x}$

② let $y_p = Ax^2 + Bx + C$

$y'_p = 2Ax + B$

$y''_p = 2A$

$2A + 6Ax + 3B + 2Ax^2 + 2Bx + 2C = x^2 + 5$

$2Ax^2 + (6A+2B)x + 2A+3B+2C = x^2+5$

$2A = 1$

$6A + 2B = 0$

$2A + 3B + 2C = 5$

$\Rightarrow \begin{cases} A = \frac{1}{2} \\ B = -\frac{3}{2} \\ C = 17 \end{cases}$ 4

$y_p = \frac{1}{2}x^2 - \frac{3}{2}x + 17$

$y = y_c + y_p$

$= c_1e^{-2x} + c_2e^{-x} + \frac{1}{2}x^2 - \frac{3}{2}x + 17$ 4

6. Solve the initial value problem $y''-4y'-12y=3e^{5t}, y(0)=\frac{18}{7}, y'(0)=\frac{-1}{7}$.

① y_c

$y_c = e^{rx}$

$y_c' = re^{rx}$

$y_c'' = r^2e^{rx}$

$r^2e^{rx} - 4re^{rx} - 12e^{rx} = 0$

$(r^2-4r-12)e^{rx} = 0$

$e^{rx}=0$ or $r^2-4r-12=0$

$(r-6)(r+2)=0$

$r_1=6$ or $r_2=-2$

$y_c = c_1e^{6x} + c_2e^{-2x}$

② y_p

$y_p = Ae^{5t}$

$y_p' = 5Ae^{5t}$

$y_p'' = 25Ae^{5t}$

$25Ae^{5t} - 20Ae^{5t} - 12Ae^{5t} = 3e^{5t}$

$-7Ae^{5t} = 3e^{5t}$

$A = -\frac{3}{7}$

$y_p = -\frac{3}{7}e^{5t}$

$y = c_1e^{6t} + c_2e^{-2t} - \frac{3}{7}e^{5t}$

$y(0) = c_1 + c_2 - \frac{3}{7} = \frac{18}{7}$

$y' = 6c_1e^{6t} - 2c_2e^{-2t} - \frac{15}{7}e^{5t}$

$y'(0) = 6c_1 - 2c_2 - \frac{15}{7} = -\frac{1}{7}$

$\Rightarrow \begin{cases} c_1 = 1 \\ c_2 = 2 \end{cases}$

$y = e^{6t} + 2e^{-2t} - \frac{3}{7}e^{5t}$

7. Evaluate the following indefinite integral:

(a) $\int x^5 + 2x\,dx$

(b) $\int \frac{3}{2x}\,dx$

(c) $\int \frac{x^5 + x + 5}{x^2}\,dx$

(d) $\int \sin^2(x)\cos^5(x)\,dx$

a) $\int x^3 + 2x\,dx = \frac{x^4}{4} + x^2 + C$

b) $\int \frac{3}{2x}\,dx = \frac{3}{2}\int \frac{1}{x}\,dx = \frac{3}{2}\ln|x| + C$

c) $\int x^3 + \frac{1}{x} + \frac{5}{x^2}\,dx = \frac{x^4}{4} + \ln|x| - 5x^{-1} + C$

d) $\int \sin^2(x)\cos^5(x)\,dx = \int \sin^2(x)\cos^4(x)\cos x\,dx$ ($\cos^5 x = \cos^2 x \cdot \cos^2 x \cdot \cos x$)

$= \int \sin^2(x)\,(1-\sin^2 x)^2\cos x\,dx$

Let $u = \sin x$; $\frac{du}{dx} = \cos x \Rightarrow dx = \frac{du}{\cos x}$

$= \int u^2(1-u^2)^2\,du$

$= \int u^2(1 - 2u^2 + u^4)\,du = \int u^2 - 2u^4 + u^6\,du$

$= \frac{u^3}{3} - \frac{2u^5}{5} + \frac{u^7}{7} + C$

$= \frac{\sin^3 x}{3} - \frac{2\sin^5 x}{5} + \frac{\sin^7 x}{7} + C$

8. Evaluate the definite integral $\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{\cos x}{\sin^2 x}\,dx$. Round your final answer to four decimal places.

Let $u = \sin x$; $\frac{du}{dx} = \cos x \Rightarrow dx = \frac{du}{\cos x}$

when $x = \frac{2\pi}{3}$, $u = \frac{\sqrt{3}}{2}$

$x = \frac{\pi}{3}$, $u = \frac{\sqrt{3}}{2}$

$= \int_{\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} \frac{1}{u^2}\,du = -\left[\frac{1}{u}\right]_{\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} = 0$

9. Use an appropriate substitution to solve the indefinite integral $\int 2x^2 e^{x^3}\,dx$.

Let $u = x^3$; $\frac{du}{dx} = 3x^2 \Rightarrow dx = \frac{du}{3x^2}$

$= \int e^u \cdot 2x^2 \cdot \frac{1}{3x^2}\,du$

$= \frac{2}{3}\int e^u\,du = \frac{2}{3}e^u + C = \frac{2}{3}e^{x^3} + C$

LIATE

10. Use an integration by parts to solve the indefinite integral $\int 2x\cos(5x)\,dx$.

A T

2 1 5

Let $u = x$: $dv = \cos(5x)\,dx$

$du = dx$ $\quad \rightarrow v = \frac{1}{5}\sin(5x)$

$= 2\cdot\frac{1}{5}\,x\cdot\sin(5x) - 2\int \frac{1}{5}\sin(5x)\,dx$

$= \frac{2}{5}x\sin(5x) + \frac{2\cos(5x)}{25} + C$

11. Consider a rational function, $f(x)=\dfrac{x+11}{x^2+x-6}$.

(a) Perform partial fraction decomposition on the given rational function.

(b) Using the partial fraction decomposition obtained in (a), evaluate the definite integral $\int_5^6 f(x)\,dx$. Round your final answer to four decimal places.

a) $f(x) = \dfrac{A}{x+3} + \dfrac{B}{x-2}$

$\dfrac{x+11}{x^2+x-6} = \dfrac{A(x-2)+B(x+3)}{(x+3)(x-2)}$

① let $x = -3$

$8 = -5A$

$A = -\frac{8}{5}$

② let $x = 2$

$13 = 5B$

$B = \frac{13}{5}$

$f(x) = -\frac{8}{5}\cdot\frac{1}{x+3} + \frac{13}{5}\cdot\frac{1}{x-2}$

b) $\int_5^6 f(x)\,dx = \int_5^6 -\frac{8}{5}\cdot\frac{1}{x+3} + \frac{13}{5}\cdot\frac{1}{x-2}\,dx$

$= -\frac{8}{5}\int_5^6 \frac{1}{x+3}\,dx + \frac{13}{5}\int_5^6 \frac{1}{x-2}\,dx$

$= -\frac{8}{5}\left[\ln|x+3|\right]_5^6 + \frac{13}{5}\left[\ln|x-2|\right]_5^6$

$= -\frac{8}{5}\left[\ln 9 - \ln 8\right] + \frac{13}{5}\left[\ln 4 - \ln 3\right]$

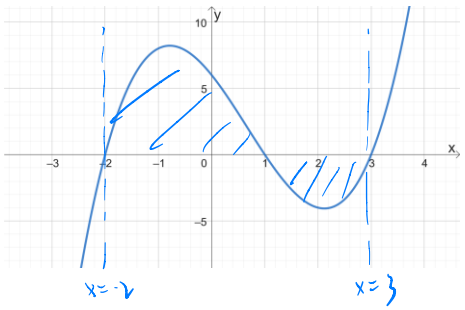
$= 0.2137 - 1.8022 = -1.5885$

$-0.1885 - 0.7479 = -0.5195$

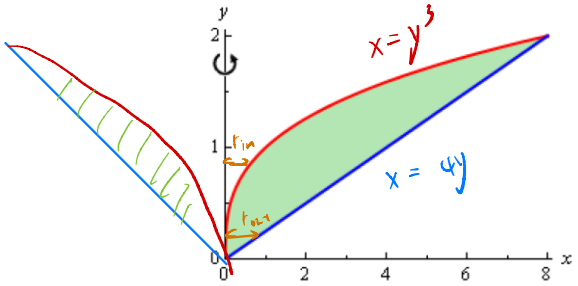
13. Use a trigonometric substitution to solve the definite integral $\int_{\frac{-4}{5}}^{\frac{-2}{5}} \frac{\sqrt{25x^2-4}}{x}\,dx$.

Round your final answer to four decimal places.

14. Figure below shows a graph of the functions $f(x)=x^3-2x^2-5x+6$. Compute the area of the region bounded by the $x-$axis, $y=f(x)$, and the lines $x=-2$ and $x=3$. Round your final answer to four decimal places.



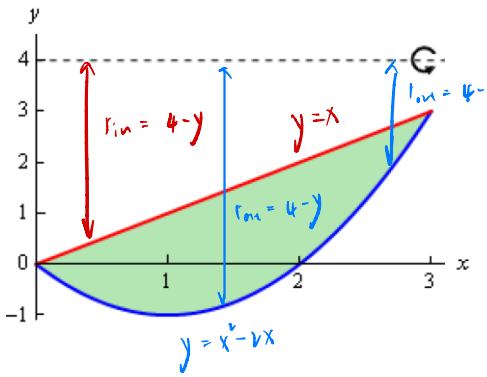
15. Figure below shows the region enclosed by the curves $y=\sqrt[3]{x}$ and $y=\frac{x}{4}$ that lies in the first quadrant. Find the volume of the solid obtained by rotating the portion of the region about the *y-axis* through Washer method.



$\int_0^2 \pi r_{out}^2 - \pi r_{in}^2\,dy$

$= \int_0^2 \pi (4y)^2 - \pi (y^3)^2\,dy$

16. Figure below shows the region enclosed by the curves $y=x^2-2x$ and $y=x$. Find the volume of the solid obtained by rotating the portion of the region about the $y=4$ through Washer method.



$A = \int_0^3 \pi r_{out}^2 - \pi r_{in}^2\,dx$

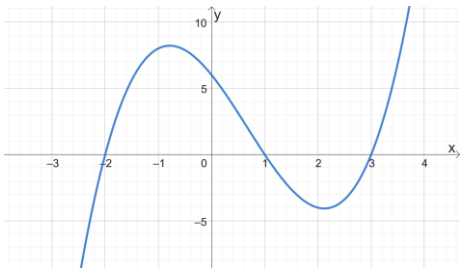
$= \int_0^3 \pi (4-x^2+2x)^2 - \pi (4-x)^2\,dx$

$\int_1^4 2\pi r\,h\,dy$

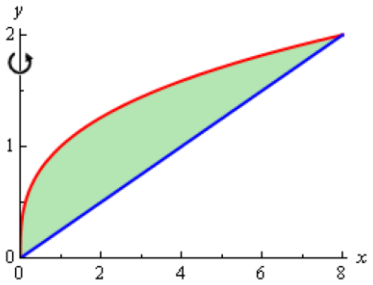
13. Use a trigonometric substitution to solve the definite integral $\int_{\frac{-4}{5}}^{\frac{-2}{5}} \frac{\sqrt{25x^2-4}}{x}\,dx$.

Round your final answer to four decimal places.

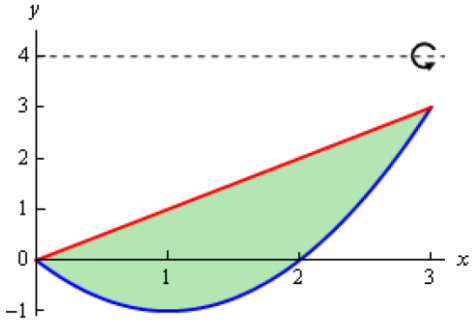
14. Figure below shows a graph of the functions $f(x)=x^3-2x^2-5x+6$. Compute the area of the region bounded by the $x-$axis, $y=f(x)$, and the lines $x=-2$ and $x=3$. Round your final answer to four decimal places.



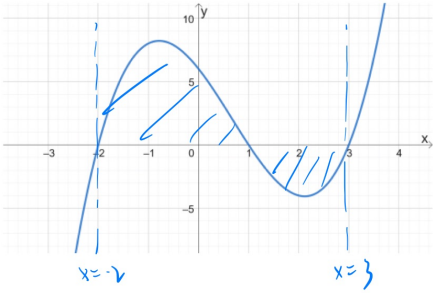
15. Figure below shows the region enclosed by the curves $y=\sqrt[3]{x}$ and $y=\frac{x}{4}$ that lies in the first quadrant. Find the volume of the solid obtained by rotating the portion of the region about the *y-axis* through Washer method.



16. Figure below shows the region enclosed by the curves $y=x^2-2x$ and $y=x$. Find the volume of the solid obtained by rotating the portion of the region about the $y=4$ through Washer method.



14. Figure below shows a graph of the functions $f(x)=x^3-2x^2-5x+6$. Compute the area of the region bounded by the $x-$axis, $y=f(x)$, and the lines $x=-2$ and $x=3$. Round your final answer to four decimal places.



$A = A_{up} - A_{low}$

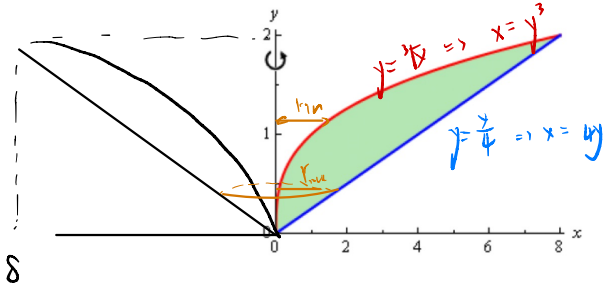
$= \int_{-2}^{1} x^3-2x^2-5x+6\,dx - \int_{1}^{3} x^3-2x^2-5x+6\,dx$

$= \left[\frac{x^4}{4} - \frac{2x^3}{3} - \frac{5x^2}{2} + 6x\right]_{-2}^{1} - \left[\frac{x^4}{4} - \frac{2x^3}{3} - \frac{5x^2}{2} + 6x\right]_{1}^{3}$

$= \frac{63}{4} + \frac{16}{3}$

$= \frac{253}{12}$

15. Figure below shows the region enclosed by the curves $y = \sqrt[3]{x}$ and $y = \frac{x}{4}$ that lies in the first quadrant. Find the volume of the solid obtained by rotating the portion of the region about the *y-axis* through Washer method.



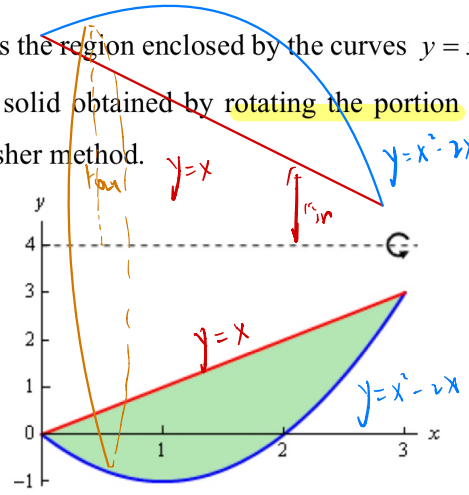
$$A = \int_0^2 \pi r_{out}^2 - \pi r_{in}^2 \, dy = \int_0^2 \pi \cdot (4y)^2 - \pi (y^3)^2 \, dy$$

$$= \int_0^2 16\pi y^2 - \pi y^6 \, dy$$

$$= \left[\frac{16}{3}\pi y^3 - \frac{\pi}{7} y^7 \right]_0^2$$

$$= \left[\frac{16}{3}\pi \cdot 2^3 - \frac{\pi}{7} \cdot 2^7 \right] = \frac{512}{21}\pi$$

16. Figure below shows the region enclosed by the curves $y = x^2 - 2x$ and $y = x$. Find the volume of the solid obtained by rotating the portion of the region about the $y = 4$ through Washer method.



$$A = \int_0^3 \pi r_{out}^2 - \pi r_{in}^2 \, dx = \int_0^3 \pi \underbrace{(x^2-2x)}_{c}^2 - \pi x^2 \, dx$$

$$(x^2-2x)^2 - 8(x^2-2x) + 16$$

$$= x^4 - 4x^3 + 4x^2 - 8x^2 + 16x + 16$$

$$= x^4 - 4x^3 - 4x^2 + 16x + 16$$

$$= \int_0^3 \pi (x^4 - 4x^3 - 4x^2 + 16x + 16 - x^2 + 8x - 16)\, dx$$

$$= \int_0^3 \pi (x^4 - 4x^3 - 5x^2 + 24x)\, dx$$

$$= \pi \left[\frac{x^5}{5} - x^4 - \frac{5}{3}x^3 + 12x^2 \right]_0^3$$

$$= \frac{153}{5}\pi$$

13. Use a trigonometric substitution to solve the definite integral $\int_{\frac{-4}{5}}^{\frac{-2}{5}} \frac{\sqrt{25x^2-4}}{x}\,dx$.

Round your final answer to four decimal places.

let $5x = 2\sec\theta \Rightarrow \theta = \cos^{-1}\left(\frac{2}{5x}\right)$

$x = \frac{2}{5}\sec\theta \;;\; dx = \frac{2}{5}\sec\theta\tan\theta\,d\theta$

when $x = -\frac{2}{5}$, $\theta = \cos^{-1}\left(-\frac{5}{4}\right) = 1.73$ (2π/3)

$x = -\frac{4}{5}$, $\theta = \cos^{-1}\left(-\frac{2}{\sqrt{4}}\right) = 1.90$ (π)

$-\frac{1}{2}$

$$= \int_{1.90}^{1.73} \frac{\sqrt{4\sec^2\theta - 4}}{\frac{2}{5}\sec\theta} \cdot \frac{2}{5}\sec\theta\tan\theta\,d\theta$$

$$= \int_{1.90}^{1.73} \frac{2\tan\theta}{\frac{2}{5}\sec\theta} \cdot \frac{2}{5}\sec\theta\tan\theta\,d\theta$$

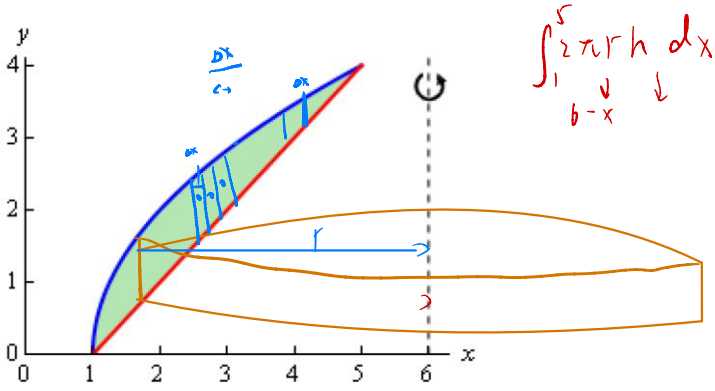
$$= 2\int_{1.90}^{1.73} \tan^2\theta\,d\theta = 2\int_{1.90}^{1.73} (\sec^2\theta - 1)\,d\theta = 2\left[\tan\theta - \theta\right]_{1.90}^{1.73}$$

$$= 2\left[\tan(1.73) - 1.73 - (\tan 1.90 - 1.90)\right]$$

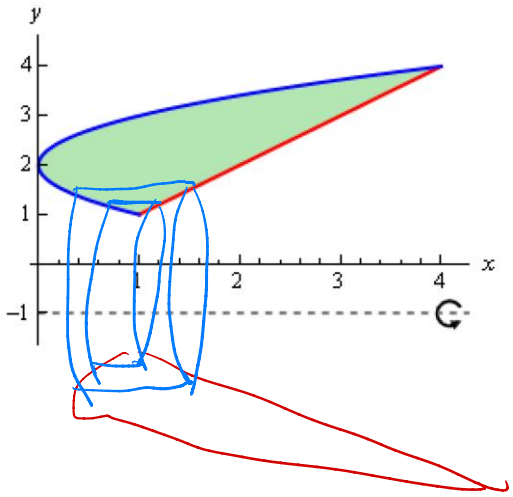
$$= 1.3697$$

Expression	Substitution	Domain	Identity
$\sqrt{a^2-x^2}$	$x = a\sin\theta$	$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$	$1-\sin^2\theta = \cos^2\theta$
$\sqrt{a^2+x^2}$	$x = a\tan\theta$	$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$	$1+\tan^2\theta = \sec^2\theta$
$\sqrt{x^2-a^2}$	$x = a\sec\theta$	$0 \le \theta \le \pi,\ \theta \ne \frac{\pi}{2}$	$\sec^2\theta - 1 = \tan^2\theta$

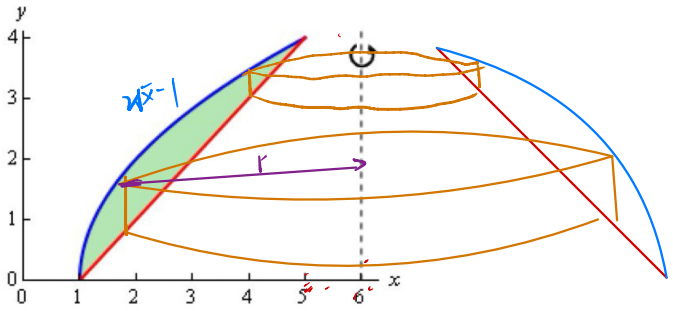
17. Figure below shows the region enclosed by the curves $y = 2\sqrt{x-1}$ and $y = x-1$. Find the volume of the solid obtained by rotating the portion of the region about the $x = 6$ through Cylindrical Shell method.



18. Figure below shows the region enclosed by the curves $x = (y-2)^2$ and $y = x$. Find the volume of the solid obtained by rotating the portion of the region about the $y = -1$ through Cylindrical Shell method.



17. Figure below shows the region enclosed by the curves $y = 2\sqrt{x-1}$ and $y = x-1$. Find the volume of the solid obtained by rotating the portion of the region about the $y = 6$ through Cylindrical Shell method.



$2\sqrt{4} - 4$

$2\sqrt{5} - 5$

$$A = \int_1^5 2\pi r\, h\, dx$$

$$= \int_1^5 2\pi (6-x)\left[(2\sqrt{x-1}) - (x-1)\right] dx$$

$$= 2\pi \int_1^5 x^2 - 7x + 6 + 12\sqrt{x-1} - 2x\sqrt{x-1}\, dx$$

$12(x-1)^{\frac{1}{2}}$

$12 \times \frac{2}{3}(x-1)^{\frac{3}{2}}$

let $u = x-1$

$$= 2\pi \left[\frac{x^3}{3} - \frac{7}{2}x^2 + 6x + 8(x-1)^{\frac{3}{2}}\right]_1^5 - 2\pi \int_1^4 2(u+1)\sqrt{u}\, du$$

$$= 2\pi\left[\frac{125}{3} - \frac{7}{2}\cdot 25 + 30 + 32 - \left(\frac{1}{3} - \frac{7}{2} + 8\right)\right] - 4\pi\left[\frac{2}{7}u^{\frac{7}{2}} + \frac{2}{3}u^{\frac{3}{2}}\right]_0^4$$

$$= \frac{284}{3}\pi - 4\pi\left[\frac{2}{7}\cdot 4^{\frac{7}{2}} + \frac{2}{3}\cdot 4^{\frac{3}{2}}\right]$$

$$= \frac{284}{3}\pi - 178.92 \qquad 37.87$$

284

227

57

$$= 57$$

$$V = \int_1^6 2\pi(6-x)\left(2\sqrt{x-1} - (x-1)\right) dx$$

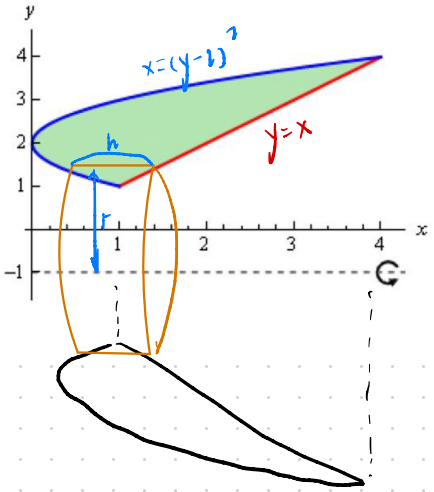
$$= 2\pi \int_1^6 x^2 - 7x + 6 + 12\sqrt{x-1} - 2x\sqrt{x-1}\, dx$$

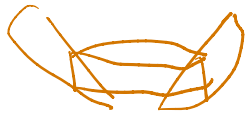
Let $u = x-1$

$$= 2\pi\left[\frac{1}{3}x^3 - \frac{7}{2}x^2 + 6x + 8(x-1)^{3/2} - \frac{4}{5}(x-1)^{\frac{5}{2}} - \frac{4}{5}(x-1)^{\frac{5}{2}}\right]_1^6$$

$$= \frac{272}{5}\pi \quad 56.828$$

18. Figure below shows the region enclosed by the curves $x=(y-2)^2$ and $y=x$. Find the volume of the solid obtained by rotating the portion of the region about the $y=-1$ through Cylindrical Shell method.





$$A = \int_1^4 2\pi r h \, dy = \int_1^4 2\pi (y+1)\left[y-(y-2)^2\right] dy$$

$$= \int_1^4 2\pi (y+1)\left[y-y^2+4y-4\right] dy$$

$$= 2\pi \int_1^4 y^2 - y^3 + 4y^2 - 4y + y - y^2 + 4y - 4 \, dy$$

$$= 2\pi \int_1^4 -y^3 + 4y^2 + y - 4 \, dy$$

$$= 2\pi \left[-\frac{y^4}{4} + \frac{4}{3}y^3 + \frac{y^2}{2} - 4y\right]_1^4$$

$$= 2\pi \left(\frac{136}{3} - \frac{11}{6}\right) \quad 2\pi\left(\frac{40}{3} + \frac{29}{12}\right)$$

$$= 181\pi \qquad = \frac{63}{2}\pi$$

COM110 202504 Final Revision — add on mixing problem

A 120-gallon tank holds purified water. Salt water with 1.5 lbs of salt per gallon leaks into the tank at a rate of 2 gallons per minute. The mixture in the tank is assumed to be constantly (perfectly) mixed, and it flows out of the tank at a rate of 3 gallon per minute. Write a model $S(t)$ for the amount of salt in the tank (measured in lbs) after t minute.

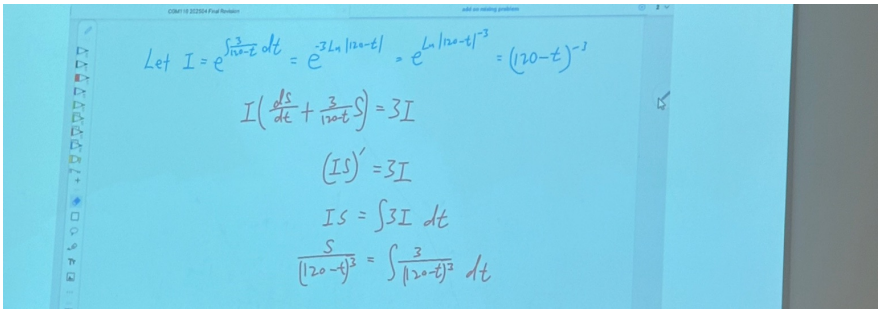
120 gallon water

$$\frac{dS}{dt} = \text{rate in} - \text{rate out}$$

$$= (1.5)(2) - (3)\left(\frac{S}{120-t}\right)$$

$$= 3 - \frac{3S}{120-t} \qquad \text{where } S(0) = 0$$

$$\frac{dS}{dt} + \frac{3S}{120-t} = 3$$



Given $S(0) = 0$

$$0 = \frac{3}{2}(120) + 120^2 c$$

$$c = \frac{-1}{9600} \qquad S = \frac{1}{2}(120-t) - \frac{(120-t)^2}{9600}$$